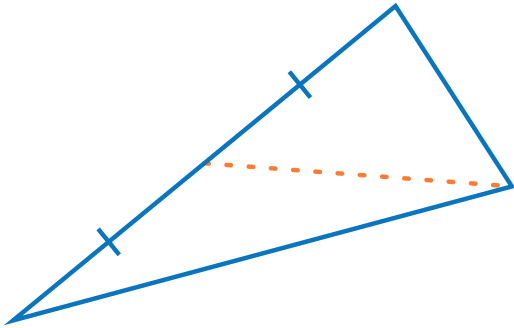


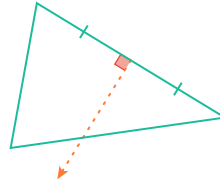
Altitudes, medians, and bisectors of triangles

1 In the triangle shown below, what does the dashed line represent?

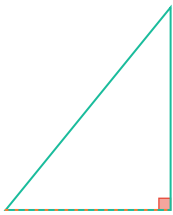
a



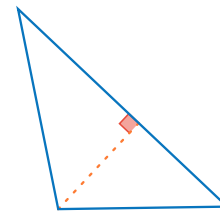
b



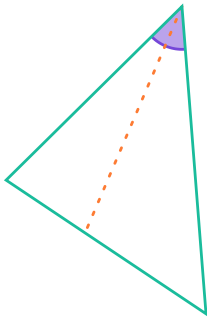
c



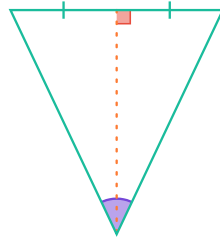
d



e

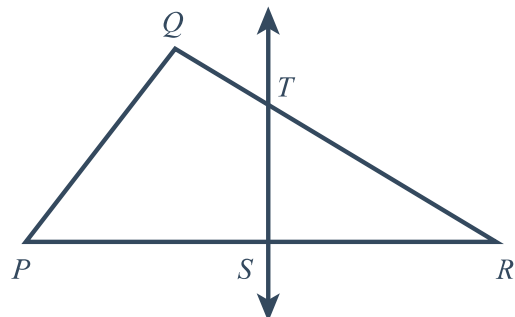


f

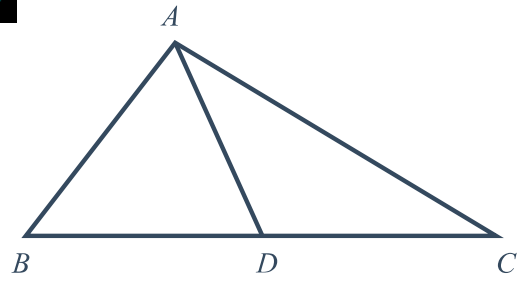


2 In $\triangle PQR$, $\overline{ST}$ is the perpendicular bisector of $\overline{PR}$.

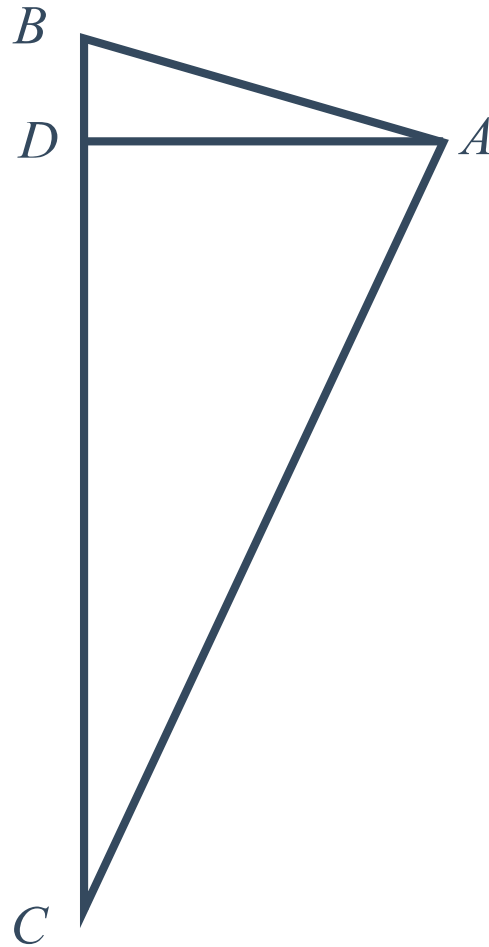
If $m\angle PST = (2x - 3)^\circ$, find x .



- 3 $\overline{AD}$ is a median of $\triangle ABC$.
If $BD = 3x - 9$, and $CD = 5x - 17$, find x .

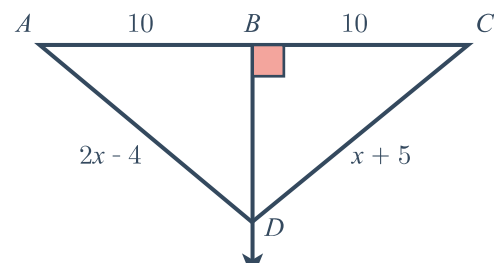


- 4 $\overline{AD}$ is an altitude of $\triangle ABC$.
If $m\angle DAC = (6x - 8)^\circ$ and
 $m\angle DCA = (4x + 13)^\circ$, find x .



- 5 Consider the following figure.

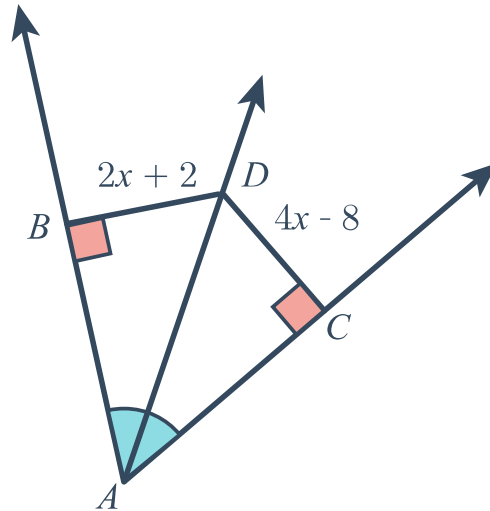
- a Solve for x .
b Find AD .





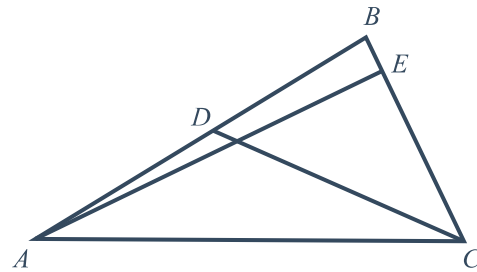
6 Consider the following diagram.

- a Solve for x .
- b Find BD .

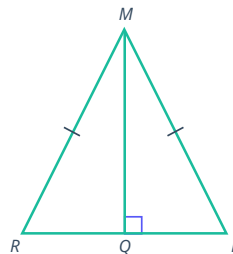


7 In $\triangle ABC$, $m\angle AEC = (3x - 6)^\circ$,
 $AD = 3y - 2$ and $BD = 5y - 8$.

- a If $\overline{AE}$ is an altitude, find x .
- b If $\overline{CD}$ is a median, find BD .



8 Given $\overline{MQ}$ is an altitude to isosceles triangle MRP , prove that $\overline{MQ}$ is also an angle bisector.



Additional questions

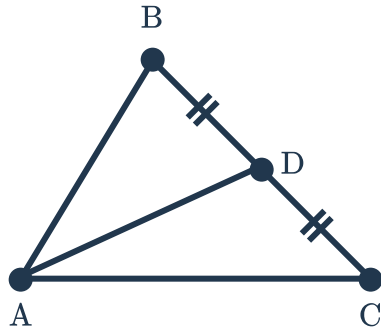
9 Write a definition for each of the following:

- a Altitude
- b Median

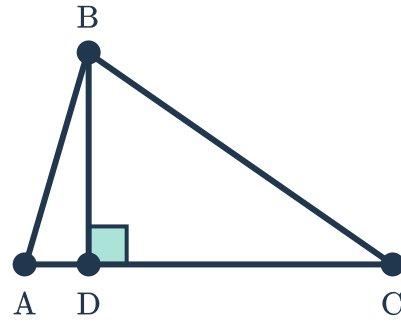
10 Identify if the stated line segment shown is median, the altitude, both, or neither in $\triangle ABC$ in the related figure.



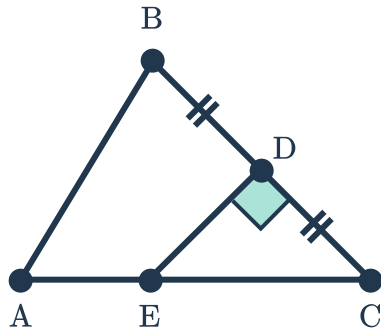
a $\overline{AD}$



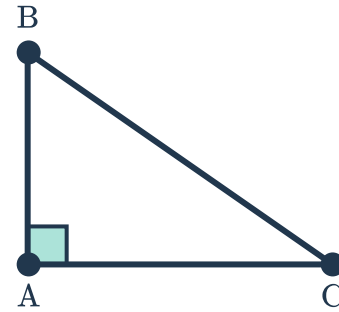
b $\overline{BD}$



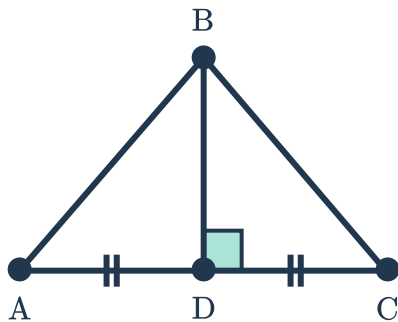
c $\overline{DE}$



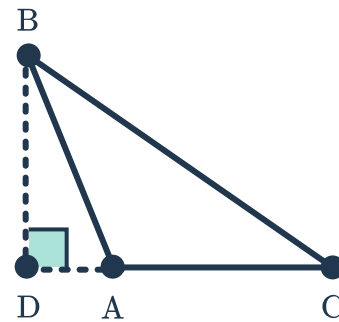
d $\overline{AB}$



e $\overline{BD}$



f $\overline{BD}$



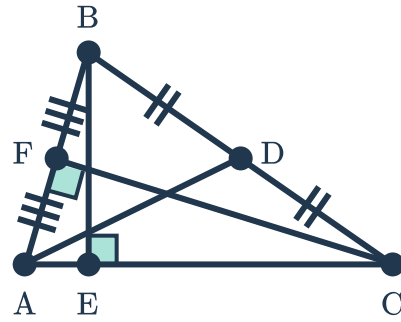
11 State the type of line segment



a $\overline{AD}$

b $\overline{BE}$

c $\overline{CF}$



12 Explain why the median to the base of an isosceles triangle is also an altitude.